Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

s

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Ale Magana	Email	coria-consortium@purdue.edu
Institute Name	Purdue University			
Address	401 Grant Street, West Lafayette, IN 47907-2045			

2. Technology Overview

Title of Proposed Technology	**Ontology-based Organizational AI Capability Diagnostic and Modular Learning Framework**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services			
	Medium Category	Small Category	Medium Category	Small Category
	Human Resource Competency Development Services	*Intelligent Learning Support / Management Technologies*		
Concept of Proposed Technology	**Core Concept** 1. [Research Motivation] Organizational Talent Alignment & Adaptive HRD Formulation • Transition to AGI Era: AI is evolving into a general-purpose technology. → Integrating AI into the working environment is necessary for survival. • Workforce Volatility & Anxiety: Rapid shifts in employment structures trigger organizational "chaos" and skill gaps. → Urgent demand for strategic workforce reallocation and targeted reskilling arises. • Imperative for "Talent Alignment": Maximizing efficiency requires bridging organizational goals and individual capabilities → Implementation of a systematic diagnosis-to-solution model is essential.			

2. [Concept of Proposed Technology]

The proposed technology is "Ontology-based Organizational AI Capability Diagnostic and Modular Learning Framework." It aligns individual competencies with organizational goals through a data-driven pipeline: Data Collection & Assessment → Assemble → Feedback Loop.

Key Functions and Potential Applications

Key Functions

To address the limitations of traditional assessment, this framework introduces four key innovations:

- [Key 1] Domain Specific Standardization: Anchors behavioral analysis to industry specific ontologies, ensuring consistent and precise interpretation of complex xAPI logs.
- [Key 2] Progress-oriented Diagnosis: Evaluates dynamic reasoning steps and behaviors, moving beyond static correctness checks.
- [Key 3] Quantitative Alignment: Quantifies the precise fit between organizational goals and individual talent using vector analysis.
- [Key 4] Continuous Evolution: Operates as a living system that self-improves its performance via a feedback loop.
- Potential Application Areas of the Proposed Technology

Potential Applications

Leveraging the ontology-driven architecture, this framework extends beyond corporate HRD to cover the entire Talent Lifecycle, establishing a universal standard for AI competency from education to employment.

- Corporate & AX Strategy (Immediate Application): Transforms subjective HR intuition into data-driven decision-making for workforce planning and AI Transformation (AX). Organizations utilize vector-based gap analysis for precise talent placement, team building, and performance prediction.
- Expansion to Education Sector (K-12 & Higher Ed):
  - University: Applies the diagnostic engine to measure "Work-Ready AI Competency" for graduates (aligned with the Purdue AI Requirement model), bridging the gap between academic knowledge and industrial demands.
  - K-12: Adapts the domain ontology to establish National AI Literacy Standards, providing foundational diagnostics for future talent cultivation at the primary and secondary education levels.
- Recruitment & Job Market Standardization: Revolutionizes the hiring market by replacing vague resumes with the HDLP (digital passport). It serves as a transparent, verified credential for skill-based Hiring, allowing recruiters to match candidates based on precise vector alignment rather than generic keywords.

Rationale for Joint R&D Project

Critical Technical Barriers and Why Global Collaboration?

This project is a strategic alliance to establish a "Global Standard AI Capability Diagnostic Framework." It leverages the unique convergence of Purdue University's institutional AI mandates and Korea's hyper-active AI adoption environment.

	1. Necessity of Global Standardization: “Why International Joint R&D?” We aim to secure a first-mover advantage by combining “theoretical rigor” with “market agility.” ● Overcoming Local Bias: AI competency standards established solely within a specific national context limit global scalability. To create a universally applicable ontology and Alignment Index, we must validate our models against diverse cultural and organizational datasets. ● Defining Global Protocols: The transition from static testing to process-based assessment is a global trend. By collaborating with a top-tier US research institution, we aim to establish the global protocols for AI literacy diagnostics and digital credentials. 2. Why Purdue University? (The “Brain” & Standard Setter) Purdue is not just a research partner; it is an active "Institutional Consumer" of this technology. ● Alignment with “AI@Purdue” Initiative: As of 2025, Purdue Trustees have approved a mandatory “AI working competency” graduation requirement. This creates an immediate, massive demand for a valid diagnostic framework to assess thousands of students. ● Theoretical Leadership: Purdue possesses world-class research capabilities in Learning Analytics and Psychometrics. They are responsible for the mathematical modeling of the core diagnostic functions (fc, ft) ensuring the theoretical validity of the metrics used for their own graduation standards. 3. The “Testbed” & Accelerator Korea is the ideal “time machine market” that reflects the future intensity of global AI usage. ● High-Density Testbed: Korea exhibits an AI usage intensity 60% higher than the US, with 20 million active users (40% of the population) and ranking 2nd globally in paid AI subscriptions. This “heavy user” environment offers the most rigorous stress-test conditions for validating diagnostic accuracy.
Key Words	1. Ontology-based Assessment 2. Process-Oriented Learning Analytics 3. Vector-based Talent Alignment 4. Self-Evolving Feedback Loop 5. Modular Learning Framework

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2)	**End of the Project**	Level (5)
R&D Objectives	**R&D Objectives** We prioritize the development of the "Diagnostic Inference Engine (The Brain)" and "Data Standard (The Language)" over peripheral features. Complex sub-tasks such as matching logic and feedback loops are integrated into the "Adaptive System Architecture" to maximize development efficiency and system cohesiveness.			

Detailed Objectives

Based on this strategy, the specific R&D objectives are:

1. Development of Industry-Agnostic AI Competency Ontology Schema ("The Universal Language")

Concept	To establish a "Master Data Standard" capable of structuring behavioral data across key representative sectors, ensuring interoperability between distinct industries.
Key Feature1	"Common Core & Industry Extension" Architecture: Design of a modular schema that separates 'Common Essentials' (shared skills like AI Ethics, Basic Prompting) from 'Industry Extensions' (specialized tasks like Coding for IT or Machine Control for Mfg).
Key Feature2	Cross-Sector Interoperability: Development of a standardized mapping logic that enables the system to adapt to major industry verticals (IT, Manufacturing, Finance) without re-engineering the core database.

2. xAPI-based Multi-dimensional Diagnostic Engines (HALI & H-AIP)

Concept	Development of dual inference engines that move beyond static testing to analyze dynamic processes and behavioral orientations.
Key Feature 1	HALI (Competency Engine): Implementation of a multi-dimensional evaluation logic utilizing prompt-based scenarios. It captures problem-solving processes, choice sequences, and revision patterns rather than single-item outcomes.
Key Feature 2	H-AIP (Propensity Engine): Design of a structured analysis framework, theoretically informed by personality diagnostics (e.g., MBTI), to evaluate directional alignment with Goal Vectors rather than judging traits as positive/negative.
Key Feature 3	Dynamic Vector Construction: Logic for constructing individual propensity vectors based on priority axes derived from leader diagnostics.

3. Vector Alignment & Adaptive Module Matching Algorithm (Assemble)

Concept	Development of the "Intelligence" that computes precise alignment gaps and prescribes personalized learning paths.
Key Feature 1	GAP Computation Logic: Quantitative measurement of the difference/distance between individual HALI/H-AIP results (HDLP_individual digital profile) and organizational Goal Vectors (Vg).
Key Feature 2	Goal Reconfiguration Structure: Design of a system that enables rapid re-diagnosis and gap recalculation in response to changes in organizational context.
Key Feature 3	Module Matching Engine: Implementation of an algorithm that calculates alignment between fixed-schema module tags and individual profiles to curate diverse learning pathways within a constrained module pool.

4. Self-Evolving Feedback Loop Architecture

Concept	Construction of a cyclical learning system that improves diagnostic accuracy and matching logic over time.
Key Feature 1	Closed-Loop Data Pipeline: Construction of an architecture that cyclically links diagnostic results, learning histories, and performance data.
Key Feature 2	Accuracy Refinement Logic: Design of a learning structure that utilizes educational outcomes to iteratively improve the diagnostic accuracy and optimize the module matching weights.

R&D Target Specifications

Core Performance Indicator	Target	Weight	Measurement & Validation Method
Test–Retest Reliability of HALI	Correlation coefficient ≥ 0.80 under repeated diagnosis in the same conditions	30%	1. Conduct the same or difficulty-matched simulation-based HALI diagnosis twice with a time interval. 2. Compute the correlation (Pearson or Spearman) between each individual's competency-axis scores across the two administrations, and validate against the overall average reliability.
Process-based Feature Dimensionality	Collect ≥ 30 quantitative, process/behavior-based features	20%	1. Define multi-dimensional behavioral characteristics (e.g., time management, revision behavior, problem-solving sequence, strategy selection, content-level signals) using xAPI/LRS-based log collection. 2. Validate based on the number of quantitative features that are practically usable for analysis.
Goal–Curriculum Alignment Score	Average alignment ≥ 70% against the organizational Goal Vector	30%	1. Using the organizational Goal Vector defined through leader diagnosis, compute alignment between the generated learning modules/curriculum and the required competency coverage via vector-based similarity (e.g., cosine similarity) or weighted coverage ratio, and evaluate the alignment score.
Performance Improvement via Feedback Loop	Confirm a monotonic increasing trend in key performance indicators across data accumulation stages	20%	1. Repeat the Diagnosis–Learning–Performance–Re-diagnosis cycle. 2. Conduct stage-wise comparative analyses of changes in key indicators (e.g., Goal–Curriculum Alignment Score, Test–Retest Reliability of HALI) as accumulated data increases (25%, 50%, 75%, 100%).

R&D Plan

1. Research Approach & Strategy

This study focuses on an organizationally aligned capability diagnosis–learning linkage framework required by changes in AI utilization environments.

Rather than adopting a single diagnostic model or fragmented instructional design, the research strategy emphasizes the construction of a cyclical, feedback loop–based framework.

- Process-oriented Assessment (HALI Engine)

  Instead of evaluating correctness of final answers alone, this approach targets problem-solving strategies, decision-making flows, and behavioral patterns as primary diagnostic objects.

  Cognitive processes revealed through prompt-based simulation scenarios are structurally interpreted as part of the diagnostic framework.

- xAPI-based Micro-Behavioral Analysis

  Beyond final responses, fine-grained behavioral logs—such as item selection order, revision history, time allocation, and decision-shift points—are utilized as core diagnostic data.

  This enables compensation for limitations inherent in traditional static assessment methods.

- Vector-based Alignment Structure

  Prior to individual diagnosis, organizational goals are mathematically defined as Goal Vectors(Vg)

  These outputs serve as reference standards for interpreting individual diagnostic results and for subsequent learning module matching.

  Individual results are vectorized into H-DLP (Humaiin Digital Literacy Profile) to serve as reference standards for precise Gap Analysis and learning module matching.

- Self-refining Feedback Loop

  By linking diagnosis, learning, and performance data, the framework is designed to progressively refine diagnostic accuracy and learning module suitability.

  This approach positions the system as an evolvable research framework, rather than a static evaluation system.

2. Key Tasks & Milestones

Year	Core Objective	Major Research Tasks	Expected Outcomes
Year 1	Foundation & Core Engine Design (Establishing the Universal Standard)	- Universal Ontology Construction: Design of the "Common Core" schema (General AI Literacy) and pilot mapping for IT & Manufacturing Basics. - Engine Prototyping (Alpha): Development of HALI Logic (Process Tracking) and H-AIP (Propensity) algorithms using prompt-based simulation scenarios. - Vector Modeling: Mathematical definition of Goal Vectors (Vg) and individual HDLP structures.	- AI Competency Ontology Schema v1.0 - Diagnostic Engine Alpha Version (Lab-level Prototype) - xAPI Data Collector Module - Alpha Pilot Test Report (N >300 Users)
Year 2	System Integration & Validation	- Deep-Tech Pilot (PoC): Deployment of the Beta system to manufacturing/construction partners - Assemble Engine Implementation: Development of the gap computation and module matching algorithm -Feedback pipeline : Collection of field performance logs	- Integrated Adaptive Learning Platform (Beta ver) - Manufacturing Extension Module - B2B PoC Performance Report (N > 3,000 Users)
Year 3	Automation & Mass Expansion	- Domain Expansion: Deployment of new domain(ex. Finance, IT,etc)modules for mass market. - Loop Automation: Full automation of the Self-Evolving Weight Optimization system. - Global Standardization: Publication of the "Global AI Literacy Protocol" with Purdue Univ.	- Commercial SaaS Launch (Official Release) - Automated Optimization Module - Service Operation (Target > 10,000+)

3. Overall Timeline

Research Activities	Year 1	Year 2	Year 3	Remarks
Design of propensity and competency diagnostic models	●			Joint

	Organizational goal and GAP modeling	●	●		Joint
	Development of H-DLP generation logic		●		Purdue-led
	Development of learning module matching structure		●	●	SME-led
	Repeated diagnostics and structural validation			●	Joint

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	1. Core development of AI-based diagnostic engines and computational structures ● Mathematical Modeling: Design and implementation of mathematical and statistical models for the HALI and H-AIP engines, ensuring scientific rigor. ● Algorithm Formalization: Formalization of core computational components, including the Goal Vector (Vg)formula and the GAP distance calculation logic, to verify convergence and stability. ● Model Verification: Conducting model-level experiments to verify the generalizability of diagnostic outcomes across different user groups. 2. Application and validation of engineering and scientific methodologies ● Scientific Review: Review and adaptation of global engineering education methodologies (e.g., IEEE, O*NET standards) to structure the Universal Competency Ontology. ● Log Interpretation Models: Development of interpretation logic for complex behavioral logs collected within the simulation environment, distinguishing between cognitive patterns and random noise. ● Statistical Frameworks: Provision of statistical frameworks to quantify abstract propensity vectors into measurable data points 3. Ensuring reliability and reproducibility of AI diagnostic systems ● Analysis of test–retest reliability, internal consistency, and sensitivity of diagnostic results under varying conditions ● Verification of model robustness across organizational, industrial, and cultural contexts ● Examination of potential causal linkages between diagnostic outcomes and educational performance ● Technical maturity with consideration for commercialization ● Consideration of computational complexity, processing speed, and scalability for real-world service deployment ● Responsibility for high-complexity modeling and validation tasks that are difficult to independently conducted by Humaiin ● Technical documentation of research outcomes and advancement of technology readiness up to the pre-commercialization stage

Expected R&R of Korean SME Partner	1. Strategic Domain Architecture & Logic Design ● HRD Logic Architecture: Leading the definition of practical "Goal Vectors" and diagnostic criteria based on field insights from 400+ corporate clients (Samsung, SK, Hyundai, etc.). ● Scenario Engineering: Designing high-fidelity Simulation Scenarios (IT/Manufacturing/Service) that serve as the foundation for the HALI diagnostic engine, utilizing real-world operational data to ensure 100% field relevance. ● Operational Constraints Definition: Defining practical requirements arising from real corporate environments (e.g., security, legacy systems) to ensure the engine is market-ready 2. Strategic PoC & Data Operations ● Managing the collection and processing of xAPI behavioral logs from a learner base of 52,000+ individuals to train the AI models with high-quality Ground Truth data. ● Collection of data reflecting diverse organizational types, job roles, and proficiency distributions ● Field-level validation of models and engines developed by Purdue University 3. Educational module and curriculum integration ● Design and provision of tagging and mapping data linking diagnostic outcomes to educational modules ● Design and validation of immediately executable curriculum composition scenarios ● Collection of post-training performance data and provision of structured feedback 4. Service-level and operational validation ● Feedback from a service perspective, including user experience (UX), operational efficiency, and management complexity ● Practical validation of diagnostic result utilization, such as workforce allocation and education planning ● Identification of requirements and refinement directions for expansion into commercial services

5. Expected Outcomes/Impact

Expected Outcomes and Impact	1. Technological Outcomes ● Establishment of an organization goal-aligned AI capability diagnostic framework ● Implementation of an integrated structure combining simulation-based high-dimensional assessment and propensity vector modeling ● Securing ontology-based H-DLP (Humaiin Digital Literacy Profile) generation technology ● Empirical validation of a self-learning feedback loop through diagnosis–education–re-diagnosis cycles 2. Commercialization Potential ● Commercialization of organization-tailored AI capability diagnosis and learning design services ● Expansion into talent assessment, reskilling programs, and workforce allocation decision-

	support tools ● Extension to individual capability history management services based on Digital Badges and Portable Credentials ● Establishment of a generalizable diagnostic framework independent of industry, country, or organizational scale 3. Industrial and Economic Impact ● Presentation of a new standard for managing AI capability through integrated qualitative–quantitative indicators ● Acceleration of the transition from education-centered AI adoption to organization- and decision-making–centered AI utilization ● Enhancement of visibility and efficiency of returns on AI investment for enterprises and public institutions ● Creation of a first-mover advantage in global AI capability diagnosis and learning design frameworks

6. References

1) Academic Research Papers

1. Swiecki, Z. (2022).
Assessment in the age of artificial intelligence.
Computers and Education: Artificial Intelligence, 3, 100075.
https://doi.org/10.1016/j.caeai.2022.100075

2. Williams, B. K. (2018).
Double-loop learning in adaptive management: The need, the challenge, and the opportunity.
Environmental Management, 62, 995–1006.
https://doi.org/10.1007/s00267-018-1107-5

3. Liu, X., et al. (2025).
The effectiveness of learning analytics-based interventions in enhancing students' learning effect: A meta-analysis of empirical studies.
SAGE Open, 15.
https://doi.org/10.1177/21582440251336707

4. Huang, C. (2025).
Enhancing learning process modeling for session-aware knowledge tracing.
Knowledge-Based Systems, 309, 112740.
https://doi.org/10.1016/j.knosys.2024.112740

5. Advanced Knowledge Tracing Research Group.
Advanced Knowledge Tracing: Incorporating Process Data and Curricula Information via an Attention-Based Framework for Accuracy and Interpretability.
(Conference paper, full text PDF)

6. Assessment Analytics Research Group. (2017).
An Ontological Model for Assessment Analytics.
(Ontology-based assessment framework paper)

7. Ontology-Driven Assessment Research Group.
Developing and validating interoperable ontology-driven game-based assessments.
Computers & Education. (ScienceDirect article)

8. Learning Analytics Community. (2016).
Learning analytics for learner awareness in remote learning environments.
(LAK Workshop paper)

9. Cooper, S. E., & Miller, L. A. (1991).
MBTI learning style–teaching style discongruencies.
Educational and Psychological Measurement, 51(3), 699–706.

---

2) Learning Analytics / Prescriptive & Diagnostic Methodologies

10. So, G.
Prescriptive Analytics (2nd Revision).
(Unpublished research manuscript)

11. Argyris, C., & Schön, D. A. (Referenced concept).
Double-loop learning theory.
(Foundational organizational learning framework)

---

3) Standards, Frameworks, and Technical Specifications

12. ADL Initiative.
Experience API (xAPI) Specification.
Advanced Distributed Learning Initiative.

13. IMS Global Learning Consortium (1EdTech).
Open Badges Specification.

14. IMS Global Learning Consortium (1EdTech).
Open Badges 3.0 Standard: Enhanced Security and Mobility.

15. IMS Global Learning Consortium (1EdTech).
Comprehensive Learner Record (CLR) Standard, Version 2.0.

---

4) Institutional Strategy & Policy References

16. Purdue University. (2025).
Purdue unveils comprehensive AI strategy; trustees approve AI working competency graduation requirement.
Purdue University Newsroom.

17. Purdue University.
AI Working Competency Graduation Requirement.
(Policy brief and public statement)

---

5) Conceptual & Technical Foundations Referenced in This Proposal

18. Learning Analytics (LA) – Process-oriented analysis of learning behaviors and decision traces
19. Process-Based Assessment (PBA) – Assessment of reasoning, decision-making, and action sequences
20. Ontology-Based Profiling – Semantic structuring of diagnostic results for consistent interpretation
21. Knowledge Tracing (KT) – Dynamic modeling of latent knowledge and behavioral states
22. Digital Badge & Portable Credential Frameworks – Decentralized, transferable records of competencies and learning history